

QueryNest Multimodal RAG - Sample Document¹

QueryNest is a multimodal document intelligence system. It ingests complex documents such as PDF, images, tables and equations, then enables hybrid retrieval and citation-based answering. The core pipeline is: parse, index, hybrid retrieval, rerank, context construction, and final answer generation.²

Hybrid retrieval in QueryNest combines Dense vector retrieval, BM25 keyword retrieval, and LightRAG knowledge-graph retrieval. All candidate hits are fused with Reciprocal Rank Fusion (RRF), deduplicated, and finally reranked before the context window is assembled.³

Method	Precision	Recall
Dense only	0.62	0.55
BM25 only	0.48	0.40
Hybrid (RRF)	0.78	0.71

⁴